

Hung Pham

SCIENTIST + TECH BUILDER • AI + SCIENCE • SCIENTIFIC SOFTWARE • QUANTUM CHEMISTRY

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Personal statement

I work at the intersection of quantum chemistry, high-performance computing, and AI, currently focused on generative modeling for chemistry and reaction-network exploration. I develop GPU-accelerated quantum chemistry software for large-scale simulation and apply deep learning and agentic systems to reaction discovery and molecular design for applications in drug discovery, materials, and energy. My development workflow is spec-first: I architect and review software systems while using AI agents to accelerate implementation, allowing me to build everything from scientific software to AI-driven products.

Experiences

Research Scientist – ByteDance (San Jose, CA)

March 2022 - Current

GENERATIVE MODELING FOR CHEMISTRY • AI FOR SCIENCE • QUANTUM CHEMISTRY • QUANTUM ALGORITHMS • PARALLEL &

GPU-ACCELERATED SOFTWARE FOR CHEMISTRY

Advisor – VBrain (Hanoi, Vietnam)

January 2026 - Current

BUILDING AN AGENTIC R&D OPERATING SYSTEM CONNECTING BUSINESSES WITH EXPERTS TO DRIVE REAL-WORLD IMPACT

Postdoctoral Research Scientist – Columbia University (New York, NY)

June 2021 - March 2022

Advisor: DAVID REICHMAN | Focus: AUXILIARY-FIELD QUANTUM MONTE-CARLO FOR PERIODIC SOLIDS

Research Assistant – University of Minnesota, Twin Cities (Minneapolis, MN)

Jan 2017 - May 2021

Advisor: LAURA GAGLIARDI | Focus: ELECTRONIC STRUCTURE THEORIES DEVELOPMENT • QUANTUM EMBEDDING THEORY •

MCSCF • STRONG CORRELATION • MATERIALS DESIGN FOR PHOTOVOLTAIC APPLICATIONS • TOPOLOGICAL INSULATORS

Teaching Assistant – University of Minnesota, Twin Cities (Minneapolis, MN)

Sep 2015 - Dec 2016

GENERAL CHEMISTRY LABS • GUEST LECTURE FOR COMPUTATIONAL CHEMISTRY (GRADUATE LEVEL)

Research Assistant – Center for Molecular and NanoArchitecture (MANAR) & Institute for Computational Science and Technology (ICST) (Ho Chi Minh, Vietnam)

Oct 2011 - Jul 2015

Advisor: NGUYEN-NGUYEN PHAM-TRAN | Focus: PERIODIC DENSITY FUNCTIONAL THEORY • GRAND CANONICAL MONTE CARLO •

RETICULAR CHEMISTRY • DESIGNING MATERIALS FOR GAS STORAGE

Education and Training

Ph.D. in Quantum Chemistry – University of Minnesota, Twin Cities

Aug 2015 - May 2021

Advisor: LAURA GAGLIARDI

B.Sc. in Chemistry – Vietnam National University Ho Chi Minh City - University of Science

Sep 2007 - Sep 2011

HONOR PROGRAM

Selected Software & Projects

FEMION — lead developer, Quantum embedding for metals, GPU random phase approximation

arXiv:2508.13036

ByteQC — contributor, GPU-accelerated quantum chemistry at scale. CUDA • MPI

129 ★

MCU — lead developer, Band structure, topology & Wannier analysis for periodic DFT

53 ★

pyWannier90 — lead developer, Wannier90 interface for VASP and PySCF

43 ★

pDMET — lead developer, Density matrix embedding theory for periodic systems (*J. Chem. Theory Comput.* 2020)

19 ★

PySCF — contributor, Core open-source quantum chemistry framework for Python

1.6k ★

Technical Skills & Software

Lead developer of: FEMION | pDMET | MCU | PyWANNIER90

Contributor of: ByteQC | PySCF | IPie

How I build: SPEC-DRIVEN & AGENTIC DEVELOPMENT • I ARCHITECT, SPECIFY, AND REVIEW; AGENTS IMPLEMENT

Codebases I own & review: PYTHON | C/C++ | CUDA | FORTRAN | BASH

AI & Machine Learning: FLOW MATCHING | SCORE-BASED & DIFFUSION MODELS | GENERATIVE MODELING | LLM APIS & AGENTIC SYSTEMS | MCP | RAG | EVALS & FINE-TUNING | PYTORCH | GNNs

HPC & Infrastructure: GPU COMPUTING | MPI | SLURM | DOCKER | GIT | CI/CD

Quantum chemistry packages: PYSCF | VASP | GAUSSIAN | CRYSTAL | QUANTUM ESPRESSO | CP2K | WANNIER90

Publications

FULL LIST OF PUBLICATIONS CAN BE FOUND IN MY **Google Scholar (clickable)** PROFILE | **Note:** * CORRESPONDING AUTHOR | †CO-FIRST AUTHOR

I. Generative Modeling & AI for Chemistry: Generative Models for Reaction Networks • Flow Matching • Score-based & Diffusion Models • Neural Wavefunctions • Graph Neural Networks • *Interpretable AI* for Catalyst Discovery

1. D. Jiang, X. Wen, Y. Chen, R. Li, W. Fu, **H. Q. Pham**, J. Chen, D. He, W. A. Goddard III, L. Wang, W. Ren, **arXiv:2508.02570**, 2025, . [🔗](#)
2. Z. Sun, C. Hu, K. Lin, J. Li, C. Cao, D. Lv, J. Chen, W. Ren, **H. Q. Pham***, **arXiv:2606.21002**, 2026, . [🔗](#)

II. Parallel & GPU-Accelerated Software: GPU Acceleration • MPI Parallelism • Tensor Optimization • High-Performance Computing • Large-Scale Scientific Simulations • Software Engineering

1. Z. Guo, Z. Huang, Q. Chen, J. Shao, G. Liu, **H. Q. Pham**, Y. Huang, C. Cao, J. Chen, D. Lv, **WIREs Comput Mol Sci**, 2025, 15. [🔗](#)
2. Y. Huang, Z. Guo, **H. Q. Pham**, D. Lv, **arXiv:2406.08314**, 2024, . [🔗](#)

III. Quantum Embedding and Quantum Algorithms: Large-scale *Ab initio* Simulations • Quantum Algorithms • HPC • Scalable Algorithms • Beyond-DFT Accuracy • 1 kcal/mol Precision • AI-Driven *ab initio* Modeling

1. C. Cao[†], **H. Q. Pham***[†], Z. Guo, Y. Zhang, Z. Huang, X. Wen, J. Chen, D. Lv, **arXiv:2508.13036**, 2025, . [🔗](#)
2. Z. Huang, Z. Guo, C. Cao, **H. Q. Pham**, X. Wen, G. H. Booth, J. Chen, D. Lv, **Nat Commun**, 2025, 16. [🔗](#)
3. C. Cao, J. Sun, X. Yuan, H.-S. Hu, **H. Q. Pham***, D. Lv, **npj Comput. Mater**, 2023, 78. [🔗](#)
4. Q. Sun, ..., **H. Q. Pham**, ..., G. K.-L. Chan, **J. Chem. Phys**, 2020, 153. [🔗](#)

IV. Quantum Monte Carlo Method Development: AFQMC • GPU-Accelerated QMC Algorithms • Strong Correlation • Embedding AFQMC • Metallic Surfaces • Efficient & Accurate *ab initio* Simulations

1. J. Lee, **H. Q. Pham**, D. R. Reichman, **J. Chem. Theory Comput.**, 2022, 18, 12, 7024 (**Front Cover, ACS Editors' Choice**). [🔗](#)

V. Computational Design of Novel Materials: DFT • Monte Carlo (MC) • Molecular Dynamics (MD) • Wave Function Methods • Tight-Binding Models • Method Development • Materials Simulation • Organic Topological Insulators • MOFs • COFs

1. **H. Q. Pham***, N.-N. Pham-Tran, **Chem. Mater.**, 2021, 33, 4488. [🔗](#)

Honors & Awards

Vietnam Education Foundation (VEF) Fellowship , VEF (declined)	2015
Grant: VNU C2013-50-03 , Vietnam National University Ho Chi Minh City	2013
Visiting Student , Omar Yaghi Lab, UCLA Department of Chemistry & Biochemistry	2011
Academic Excellent Scholarship , Viet Nam National University Ho Chi Minh City - University of Science	2008-2011
Lawrence S. Ting Scholarship , Lawrence S. Ting Memorial Fund	2008
3rd Place (Shared) , Vietnam National Olympiad in Chemistry for High-School Students	2007

Review Service

Journal list: 1. JOURNAL OF CHEMICAL THEORY AND COMPUTATION, ACS | 2. JOURNAL OF CHEMICAL PHYSICS, AIP | 3. PHYSICAL CHEMISTRY CHEMICAL PHYSICS, RSC | 4. IEEE TRANSACTIONS ON NANOTECHNOLOGY | 5. SOLID STATE COMMUNICATIONS, ELSEVIER | 6. NEW JOURNAL OF CHEMISTRY, RSC | 7. JOURNAL OF OPEN SOURCE SOFTWARE. | 8. FUTURE DRUG DISCOVERY, TAYLOR & FRANCIS.